

THILAK MURUGAN RAJSEKAR

Personal and Professional Knowledge Base for Thilak's AI Buddy

PERSONAL BACKGROUND

Full Name: Thilak Murugan Rajsekar

Birthplace: Coimbatore, Tamil Nadu, India

Upbringing: Thilak was born in Coimbatore, Tamil Nadu, India, and was brought up in both Coimbatore and Bangalore, India.

Current Location: San Jose, California, United States

Email: thilakmurugan5005@gmail.com

Phone: 321-977-3234

Sports: Thilak plays football and cricket. He has played football at the state level and cricket at the district level.

QUICK PERSONAL FACTS

- If asked where Thilak was born: Thilak was born in Coimbatore, Tamil Nadu, India.
- If asked where Thilak grew up or was brought up: Thilak was brought up in both Coimbatore and Bangalore, India.
- Thilak is a sports person who plays football and cricket.
- Thilak has played football at the state level and cricket at the district level.

PROFESSIONAL PROFILE

Senior Machine Learning Engineer and AI Innovator with 5+ years of experience building and deploying production-grade Agentic AI, Enterprise AI, and AI-native applications. Proven expertise designing stateful agent workflows, tool calling, agent orchestration, RAG, structured outputs, failure recovery, LLM evaluation, and human-in-the-loop automation. Strong hands-on experience integrating LLMs with the ServiceNow Platform, enterprise data, and downstream systems using Python, LangGraph, LangChain, Databricks, Now Assist, OpenAI/Azure OpenAI, and REST APIs. Experienced in delivering scalable and reliable AI solutions with emphasis on observability, groundedness, security, and production quality.

TECHNICAL SKILLS

Agentic AI and LLMs: Agentic AI, LLM Agents, Tool Calling, Agent Orchestration, Multi-Agent Workflows, Prompt Engineering, Structured Outputs, Conversational Memory, Human-in-the-Loop AI, LLM Evaluation

Generative AI Search and NLP: RAG, Information Retrieval, Semantic Search, Vector Search, Embeddings, Query Understanding, Retrieval, Ranking, Reranking, Search Relevance, Question Answering, NLU/NLG, Text Summarization

Machine Learning and Data: Predictive Modeling, Recommendation Systems, Feature Engineering, Model Evaluation, Similarity Matching, A/B Testing, Data Quality, Data/Model Drift, ETL Pipelines, Production Monitoring

Programming and Frameworks: Python, SQL, REST APIs, LangChain, LangGraph, JavaScript, Java, JSON, XML, LlamaIndex, FAISS, ChromaDB, LangSmith, Streamlit, scikit-learn

Cloud, Enterprise and MLOps: Databricks, Snowflake, Azure OpenAI/OpenAI, ServiceNow, Moveworks Creator Studio, SAP/ARIBA, Docker, Git, Jenkins, Azure MLOps, RBAC, Secure API Integrations

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer - AI Innovator | ServiceNow Enterprise AI Team | Nov 2024 - Present

- Designed and deployed production-grade Agentic AI and AI-native enterprise applications using LLM reasoning, enterprise retrieval, tool calling, orchestration, validation, and human-in-the-loop workflows across Finance, Support, and platform use cases.
- Built stateful agent workflows over ServiceNow records and enterprise data, invoking tools and REST APIs, validating outputs, and escalating actions, reducing manual effort by 50-70%.
- Developed LangGraph-based orchestration with conditional routing, shared state, tool-gating, structured JSON outputs, retries, fallbacks, and failure recovery for reliable production workflows.
- Integrated AI solutions with ServiceNow Platform, Now Assist, Creator Studio, Databricks, SAP/ARIBA, enterprise knowledge, and secure REST APIs.
- Implemented LLM evaluation, retrieval validation, groundedness checks, RBAC, observability, and human escalation, reducing manual review effort by 50%+.
- Delivered multiple Enterprise AI solutions that reduced triage, support, and finance-processing effort by up to 60-70%.

AI Triage Agent for Enterprise Case Workflows

- Built and deployed an AI Triage Agent on Databricks Lakehouse Monitoring to detect data/model drift, data-quality issues, pipeline anomalies, and operational risks.
- Designed a multi-step agentic investigation workflow for anomaly analysis, root-cause identification, severity classification, and remediation recommendations.
- Implemented Model Context Protocol (MCP) interfaces for standardized access to monitoring tools, enterprise data, and downstream services.
- Designed Agent-to-Agent (A2A) coordination between triage, retrieval, and remediation agents using structured messages and shared workflow state.
- Integrated with Moveworks Creator Studio for conversational investigation, follow-up reasoning, and grounded monitoring insights.
- Added structured outputs, evaluation, fallback handling, human escalation, RBAC, and AI observability for production reliability.
- Reduced manual triage and investigation effort by 60-70% through automated analysis and remediation guidance.

Agentic AI Case Intelligence and Finance Automation

- Architected an Agentic AI workflow for remittance analysis, invoice extraction, SAP/ARIBA retrieval, payment validation, and clearing-date prediction.
- Built tool-gated workflows combining LLM reasoning, enterprise retrieval, structured extraction, validation, deterministic business rules, and exception handling.
- Automated extraction and validation across PDFs, emails, ServiceNow cases, and transaction data, improving data-capture consistency by 50%+.
- Reduced manual finance review by 60-70% and improved processing efficiency by 50-60% through automated retrieval, validation, and prediction.

AI-Driven Technical Support Automation Agent

- Built an Agentic AI support agent using RAG, semantic search, ranking/reranking, and enterprise knowledge retrieval to recommend grounded incident resolutions.
- Implemented LLM evaluation, response validation, structured outputs, fallback logic, and human escalation for reliable customer-facing recommendations.
- Added retrieval relevance, completeness, and groundedness checks before generating resolution recommendations.
- Reduced manual support triage by 50-60% and first-response time by 60%+ through automated retrieval and resolution workflows.

AI/ML Engineer | University of Central Florida | Oct 2023 - Nov 2024

- Developed an AI chatbot using OpenAI GPT-4o, Llama-2, embeddings, and ChromaDB for contextual enterprise question answering.

- Implemented advanced Retrieval-Augmented Generation (RAG) with Document Summary Indexing, improving conversational accuracy by 70%, and deployed using Snowflake Cortex.
- Applied AutoML, feature engineering, and prompt engineering to improve model development and AI-driven decision-making.
- Used LangSmith for LLM tracing, evaluation, and conversation-quality monitoring, increasing query resolution by 25%.
- Built ETL pipelines, workflow automation, and application integrations to support scalable AI/data workflows.
- Deployed the chatbot POC using Streamlit and applied Git best practices for version control and collaboration.

Machine Learning Teaching Assistant | University of Central Florida | Jan 2023 - Aug 2023

- Instructed and mentored more than 40 Graduate Data Analytics students in creating machine learning projects from scratch using Python.

Machine Learning Engineer | Covalensedigital | July 2019 - Nov 2021

- Architected end-to-end ML deployment pipelines using Azure AutoML, Snowflake, Git, Docker, Azure MLOps, and Jenkins, reducing model deployment time by 42%.
- Created 50+ ETL pipelines to process and transform 10M+ customer streaming records using Snowflake, Azure Data Factory, and Azure Synapse Analytics for scalable data warehousing and analytics.
- Developed a recommendation system using SVD and Wide & Deep models with automated feature engineering, improving personalized product suggestions and increasing revenue by 30%.
- Optimized SQL queries through indexing, partitioning, and query plan analysis, reducing query execution time by 30%.
- Improved model performance through EDA, feature engineering, hyperparameter tuning, and A/B testing, increasing segmentation accuracy by 25% and enhancing system efficiency by 20%.

Data Integration Analyst | Deloitte | Feb 2019 - Jul 2019

- Integrated banking and financial data from 10+ sources using SAP CPI and REST APIs.
- Designed and monitored 100+ data pipelines and executed 50+ secure ETL workflows across on-premises and cloud systems.

PROJECT EXPERIENCE

Taxation-AI

- Developed and deployed Taxation-AI using Generative AI to automate tax calculations from invoices, reducing manual effort by 70% and providing real-time analysis through a Streamlit interface.

Document AI Chatbot (Document Query)

- Developed a chatbot allowing users to upload PDFs and ask questions, leveraging GPT-3.5 Turbo and FAISS for efficient semantic search and retrieval using OpenAI embeddings.
- Implemented RAG, reranking, and memory to improve response accuracy and user experience by 20%.

Personalized Conversational AI Chatbot

- Developed and deployed an AI chatbot using GPT-3.5 Turbo, OpenAI embeddings, FAISS, LangChain, and Streamlit.
- Achieved 85% user satisfaction and 30% improved contextual accuracy through Advanced RAG methodology and LangSmith-based evaluation.

CERTIFICATIONS

- Generative AI Professional (1Z0-1127-24) - Oracle Certificate - 2024
- Microsoft Azure Data Fundamentals - 2023
- SAP Cloud Platform Integration Certificate - SAP - 2020

EDUCATION

- Master of Science in Data Analytics | University of Central Florida | Aug 2021 - Aug 2023
- Bachelor's in Computer Science | PSG College of Technology | Jul 2016 - Jun 2020

KNOWLEDGE BASE USAGE NOTE

This document consolidates Thilak Murugan Rajsekar's personal background and the professional information in his latest uploaded resume. For chatbot questions, use the explicit personal facts above for birthplace, upbringing, location, and sports, and use the professional sections for work experience, projects, skills, certifications, and education.